

Сингулярные возмущения. Собственные частоты

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In[404]:=

```
ClearAll["Global`*"]  
$Assumptions = Element[n, PositiveIntegers]
```

Out[405]=

$n \in \mathbb{Z} \ \&\& \ n > 0$

Первая краевая задача (жесткая заделка с двух сторон)

In[406]:=

```
Eq = -μ^2 * D[X[x], {x, 4}] + D[X[x], {x, 2}] + Δ * X[x] == 0
```

Out[406]=

$\Delta X[x] + X''[x] - \mu^2 X^{(4)}[x] == 0$

In[407]:=

```
Border[f_] := {f /. x -> 0, f /. x -> 1, D[f, x] /. x -> 0, D[f, x] /. x -> 1};
```

Ищем приближенное решение

In[408]:=

```
sX = X[0][x] + μ * X[1][x] + μ^2 * X[2][x]
```

Out[408]=

$X[0][x] + \mu X[1][x] + \mu^2 X[2][x]$

In[409]:=

```
sΔ = Δ[0] + μ * Δ[1] + μ ^ 2 * Δ[2]
```

Out[409]=

$$\Delta[0] + \mu \Delta[1] + \mu^2 \Delta[2]$$

In[410]:=

```
SelfEq = Thread[CoefficientList[Eq[[1]] /. Join[Table[D[X[x] → sX, {x, i}], {i, 0, 4}], {Δ → sΔ}], μ] == 0]
```

Out[410]=

$$\left\{ \Delta[0] \times X[0][x] + X[0]''[x] == 0, \Delta[1] \times X[0][x] + \Delta[0] \times X[1][x] + X[1]''[x] == 0, \right. \\ \Delta[2] \times X[0][x] + \Delta[1] \times X[1][x] + \Delta[0] \times X[2][x] + X[2]''[x] - X[0]^{(4)}[x] == 0, \\ \left. \Delta[2] \times X[1][x] + \Delta[1] \times X[2][x] - X[1]^{(4)}[x] == 0, \Delta[2] \times X[2][x] - X[2]^{(4)}[x] == 0 \right\}$$

In[411]:=

```
SolX0 = DSolve[{SelfEq[[1]]}, X[0][x], x][[1]] /. {C[2] → 0, C[1] → 1}
```

Out[411]=

$$\{X[0][x] \rightarrow \sin[x \sqrt{\Delta[0]}]\}$$

In[412]:=

```
SolX1 = Simplify[DSolve[Simplify[SelfEq[[2]] /. SolX0], X[1][x], x][[1]]]
```

Out[412]=

$$\left\{ X[1][x] \rightarrow \frac{\sin[x \sqrt{\Delta[0]}] (4 c_1 \Delta[0] - \Delta[1]) + 2 \cos[x \sqrt{\Delta[0]}] (2 c_2 \Delta[0] + x \sqrt{\Delta[0]} \Delta[1])}{4 \Delta[0]} \right\}$$

In[413]:=

```
SolX2 = Simplify[DSolve[SelfEq[[3]] /. Join[SolX0, SolX1, D[SolX0, {x, 4}]], X[2][x], x][[1]]]
```

Out[413]=

$$\left\{ X[2][x] \rightarrow \frac{1}{8 \Delta[0]^2} \left(\sin[x \sqrt{\Delta[0]}] \left(8 c_3 \Delta[0]^2 + 2 \Delta[0]^3 - 4 x c_2 \Delta[0]^{3/2} \Delta[1] + \Delta[1]^2 - \Delta[0] (2 c_1 \Delta[1] + x^2 \Delta[1]^2 + 2 \Delta[2]) \right) - \right. \right. \\ \left. \left. 2 \cos[x \sqrt{\Delta[0]}] \sqrt{\Delta[0]} (-4 c_4 \Delta[0]^{3/2} + c_2 \sqrt{\Delta[0]} \Delta[1] + x (2 \Delta[0]^3 + \Delta[1]^2 - 2 \Delta[0] (c_1 \Delta[1] + \Delta[2]))) \right) \right\}$$

0-е приближение

In[414]:=

```
SolX0a = {X[0][x] → SolX0[1, 2] + μ * (F[1] * Exp[-x / μ] + F[2] * Exp[(x - 1) / μ])}
```

Out[414]=

$$\left\{ X[0][x] \rightarrow \mu \left(e^{-\frac{x}{\mu}} F[1] + e^{\frac{-1+x}{\mu}} F[2] \right) + \sin\left[x \sqrt{\Lambda[0]}\right] \right\}$$

In[415]:=

```
SolC0a = Solve[Thread[Limit[Coefficient[Border[SolX0a[1, 2]][[3;;4]], μ, 0], μ → 0, Direction → "FromAbove"] == 0], {F[1], F[2]}][[1]]
```

Out[415]=

$$\left\{ F[1] \rightarrow \sqrt{\Lambda[0]}, F[2] \rightarrow -\cos\left[\sqrt{\Lambda[0]}\right] \sqrt{\Lambda[0]} \right\}$$

In[416]:=

```
SolΛ0a = {Λ[0] → (Pi * n) ^ 2}
```

Out[416]=

$$\left\{ \Lambda[0] \rightarrow n^2 \pi^2 \right\}$$

In[417]:=

```
SolX0a = Simplify[SolX0a /. SolC0a /. SolΛ0a]
```

Out[417]=

$$\left\{ X[0][x] \rightarrow -(-1)^n e^{\frac{-1+x}{\mu}} n \pi \mu + e^{-\frac{x}{\mu}} n \pi \mu + \sin[n \pi x] \right\}$$

In[418]:=

```
Simplify[Border[SolX0a[1, 2]]]
```

Out[418]=

$$\left\{ \left(1 - (-1)^n e^{-1/\mu}\right) n \pi \mu, \left(-(-1)^n + e^{-1/\mu}\right) n \pi \mu, -(-1)^n e^{-1/\mu} n \pi, -e^{-1/\mu} n \pi \right\}$$

1-е приближение

In[419]:=

```
SolX1a = Simplify[SolX1 /. SolA0a]
```

Out[419]=

$$\left\{ X[1][x] \rightarrow \frac{\sin[n\pi x] (4n^2\pi^2 c_1 - \Delta[1]) + 2n\pi \cos[n\pi x] (2n\pi c_2 + x\Delta[1])}{4n^2\pi^2} \right\}$$

In[420]:=

```
SolC1a = Solve[Thread[Limit[Coefficient[Border[SolX0a[[1, 2]] + μ * SolX1a[[1, 2]]], μ, 1], μ → 0, Direction → "FromAbove"] == 0],
{C[1], C[2], Δ[1]}][[1]]
```

Out[420]=

$$\{c_1 \rightarrow -1, c_2 \rightarrow -n\pi, \Delta[1] \rightarrow 4n^2\pi^2\}$$

In[421]:=

```
SolA1a = {SolC1a[[3]]}
```

Out[421]=

$$\{\Delta[1] \rightarrow 4n^2\pi^2\}$$

In[422]:=

```
SolX1a = Simplify[SolX1a /. SolC1a]
```

Out[422]=

$$\{X[1][x] \rightarrow n\pi(-1 + 2x)\cos[n\pi x] - 2\sin[n\pi x]\}$$

In[423]:=

```
Simplify[Border[SolX0a[[1, 2]] + μ * SolX1a[[1, 2]]]
```

Out[423]=

$$\{-(-1)^n e^{-1/\mu} n\pi\mu, e^{-1/\mu} n\pi\mu, -(-1)^n e^{-1/\mu} n\pi, -e^{-1/\mu} n\pi\}$$

Заметим, что искать поправку к первому приближению не нужно, так оно само с поправкой нулевого приближения удовлетворяет краевым условиям с экспоненциально высокой точностью.

2-е приближение

In[424]:=

```
SolX2a = Simplify[SolX2 /. SolC1a /. SolA0a /. SolA1a]
```

Out[424]=

$$\left\{ X[2][x] \rightarrow \frac{\sin[n\pi x] \left(n^4 \pi^4 (1 + 8x - 8x^2) + 4n^2 \pi^2 (3 + c_3) - \Lambda[2] \right) - 2n\pi \cos[n\pi x] \left(n^4 \pi^4 x + 2n^2 \pi^2 (-1 + 6x) - 2n\pi c_4 - x \Lambda[2] \right)}{4n^2 \pi^2} \right\}$$

In[425]:=

```
eq = Simplify[Thread[Coefficient[Border[SolX0a[[1, 2]] + μ * SolX1a[[1, 2]] + μ^2 * SolX2a[[1, 2]], μ, 2] == 0]]
```

Out[425]=

$$\left\{ n\pi + c_4 == 0, 10n^2 \pi^2 + n^4 \pi^4 == 2n\pi c_4 + \Lambda[2], n^4 \pi^4 == 4n^2 \pi^2 (-3 + c_3) + \Lambda[2], n^4 \pi^4 == 4n^2 \pi^2 (-3 + c_3) + \Lambda[2] \right\}$$

In[426]:=

```
SolC2a = Solve[Simplify[Thread[Coefficient[Border[SolX0a[[1, 2]] + μ * SolX1a[[1, 2]] + μ^2 * SolX2a[[1, 2]], μ, 2] == 0]], {C[3], C[4], Λ[2]}][[1]]
```

... Solve: Equations may not give solutions for all "solve" variables.

Out[426]=

$$\left\{ c_3 \rightarrow 0, c_4 \rightarrow -n\pi, \Lambda[2] \rightarrow \pi^2 (12n^2 + n^4 \pi^2) \right\}$$

In[427]:=

```
SolA2a = {SolC2a[[3]]}
```

Out[427]=

$$\left\{ \Lambda[2] \rightarrow \pi^2 (12n^2 + n^4 \pi^2) \right\}$$

In[428]:=

```
SolX2a = Simplify[SolX2a /. SolC2a]
```

Out[428]=

$$\left\{ X[2][x] \rightarrow -2n^2 \pi^2 (-1 + x) x \sin[n\pi x] \right\}$$

In[429]:=

```
Simplify[Border[SolX0a[[1, 2]] + μ * SolX1a[[1, 2]] + μ^2 * SolX2a[[1, 2]]]
```

Out[429]=

$$\left\{ -(-1)^n e^{-1/\mu} n \pi \mu, e^{-1/\mu} n \pi \mu, -(-1)^n e^{-1/\mu} n \pi, -e^{-1/\mu} n \pi \right\}$$

Заметим, что искать поправку ко второму приближению тоже не нужно, так оно само с поправкой нулевого приближения удовлетворяет краевым условиям с экспоненциально высокой точностью.

Ответ

In[430]:=

```
Print["Λ(μ) = ", sΛ /. Join[SolΛ0a, SolΛ1a, SolΛ2a]]
Print["X(μ) = ", Simplify[sX /. Join[SolX0a, SolX1a, SolX2a]]]
```

$$\Lambda(\mu) = n^2 \pi^2 + 4 n^2 \pi^2 \mu + \pi^2 (12 n^2 + n^4 \pi^2) \mu^2$$

$$X(\mu) = e^{-\frac{1+x}{\mu}} \left(e^{\frac{1}{\mu}} - (-1)^n e^{\frac{2x}{\mu}} \right) n \pi \mu + n \pi (-1 + 2x) \mu \cos[n \pi x] - (-1 + 2\mu + 2 n^2 \pi^2 (-1 + x) x \mu^2) \sin[n \pi x]$$

Вторая краевая задача (жесткая заделка слева, шарнирное опирание справа)

In[432]:=

```
Border[f_] := {f /. x → 0, f /. x → 1, D[f, x] /. x → 0, D[f, {x, 2}] /. x → 1};
```

0-е приближение

In[433]:=

```
SolX0b = {X[0][x] → SolX0[[1, 2]] + μ * F[1] * Exp[-x / μ] + μ^2 * F[2] * Exp[(x - 1) / μ]}
```

Out[433]=

$$\left\{ X[0][x] \rightarrow e^{-\frac{x}{\mu}} \mu F[1] + e^{\frac{-1+x}{\mu}} \mu^2 F[2] + \sin\left[x \sqrt{\Lambda[0]}\right] \right\}$$

In[434]:=

```
SolC0b = Solve[Thread[Limit[Coefficient[Border[SolX0b[[1, 2]]][3;;4], μ, 0], μ → 0, Direction → "FromAbove"] == 0], {F[1], F[2]}}][[1]]
```

Out[434]=

$$\{F[1] \rightarrow \sqrt{\Delta[0]}, F[2] \rightarrow \sin[\sqrt{\Delta[0]}] \Delta[0]\}$$

In[435]:=

```
SolA0b = {Δ[0] → (Pi * n) ^ 2}
```

Out[435]=

$$\{\Delta[0] \rightarrow n^2 \pi^2\}$$

In[436]:=

```
SolX0b = Simplify[SolX0b /. SolC0b /. SolA0b]
```

Out[436]=

$$\{X[0][x] \rightarrow e^{-\frac{x}{\mu}} n \pi \mu + \sin[n \pi x]\}$$

In[437]:=

```
Simplify[Border[SolX0b[[1, 2]]]
```

Out[437]=

$$\{n \pi \mu, e^{-1/\mu} n \pi \mu, 0, \frac{e^{-1/\mu} n \pi}{\mu}\}$$

1-е приближение

In[438]:=

```
SolX1b = Simplify[{X[1][x] → SolX1[[1, 2]] /. SolA0b}]
```

Out[438]=

$$\{X[1][x] \rightarrow \frac{\sin[n \pi x] (4 n^2 \pi^2 c_1 - \Delta[1]) + 2 n \pi \cos[n \pi x] (2 n \pi c_2 + x \Delta[1])}{4 n^2 \pi^2}\}$$

In[439]:=

```
SolC1b = Solve[Thread[Limit[Coefficient[Border[SolX0b[[1, 2]] +  $\mu$  * SolX1b[[1, 2]]],  $\mu$ , 1],  $\mu \rightarrow 0$ , Direction  $\rightarrow$  "FromAbove"] == 0],
  {C[1], C[2],  $\Delta$ [1]}][[1]]
```

Out[439]=

$$\left\{ c_1 \rightarrow -\frac{1}{2}, c_2 \rightarrow -n\pi, \Delta[1] \rightarrow 2n^2\pi^2 \right\}$$

In[440]:=

```
Sol $\Delta$ 1b = {SolC1b[[3]]}
```

Out[440]=

$$\{\Delta[1] \rightarrow 2n^2\pi^2\}$$

In[441]:=

```
SolX1b = Simplify[SolX1b /. SolC1b]
```

Out[441]=

$$\{X[1][x] \rightarrow n\pi(-1+x)\cos[n\pi x] - \sin[n\pi x]\}$$

In[442]:=

```
Simplify[Border[SolX0b[[1, 2]] +  $\mu$  * SolX1b[[1, 2]]]
```

Out[442]=

$$\left\{ 0, e^{-1/\mu} n\pi\mu, 0, \frac{e^{-1/\mu} n\pi}{\mu} \right\}$$

Заметим, что искать поправку к первому приближению не нужно, так оно само с поправкой нулевого приближения удовлетворяет краевым условиям с экспоненциально высокой точностью или абсолютно точно.

2-е приближение

In[443]:=

```
SolX2b = Simplify[{X[2][x] → SolX2[[1, 2]] /. SolC1b /. SolA0b}]
```

Out[443]=

$$\left\{ X[2][x] \rightarrow \frac{\sin[n\pi x] \left(n^4 \pi^4 (1 + 4x - 2x^2) + n^2 \pi^2 (3 + 4c_3) - \Delta[2] \right) - 2n\pi \cos[n\pi x] \left(n^4 \pi^4 x + n^2 \pi^2 (-1 + 3x) - 2n\pi c_4 - x \Delta[2] \right)}{4n^2 \pi^2} \right\}$$

In[444]:=

```
SolC2b = Solve[Thread[Limit[Coefficient[Simplify[Border[SolX0b[[1, 2]] + μ * SolX1b[[1, 2]] + μ^2 * SolX2b[[1, 2]]], μ, 2],  
μ → 0, Direction → "FromAbove"] == 0], {C[3], C[4], Δ[2]}][[1]]
```

Out[444]=

$$\left\{ c_3 \rightarrow 0, c_4 \rightarrow -\frac{n\pi}{2}, \Delta[2] \rightarrow \pi^2 (3n^2 + n^4 \pi^2) \right\}$$

In[445]:=

```
SolA2b = {SolC2b[[3]]}
```

Out[445]=

$$\left\{ \Delta[2] \rightarrow \pi^2 (3n^2 + n^4 \pi^2) \right\}$$

In[446]:=

```
SolX2b = Simplify[SolX2b /. SolC2b]
```

Out[446]=

$$\left\{ X[2][x] \rightarrow -\frac{1}{2} n^2 \pi^2 (-2 + x) x \sin[n\pi x] \right\}$$

In[447]:=

```
Simplify[Border[SolX0b[[1, 2]] + μ * SolX1b[[1, 2]] + μ^2 * SolX2b[[1, 2]]]
```

Out[447]=

$$\left\{ 0, e^{-1/\mu} n\pi\mu, 0, \frac{e^{-1/\mu} n\pi}{\mu} \right\}$$

Заметим, что второе краевое условие теперь выполнено с точностью до μ^4 , остальные условия выполнены либо точно, либо с точностью до экспоненциального члена. Без краевой поправки второго порядка четвертое краевое условие было бы выполнено с точностью до μ^2 , а остальные либо абсолютно точно, либо с точностью до экспоненциального члена.

Ответ

In[448]:=

```
Print["Λ(μ) = ", sΛ /. Join[SolΛ0b, SolΛ1b, SolΛ2b]]
Print["X(μ) = ", Simplify[sX /. Join[SolX0b, SolX1b, SolX2b]]]
```

$$\Lambda(\mu) = n^2 \pi^2 + 2 n^2 \pi^2 \mu + \pi^2 (3 n^2 + n^4 \pi^2) \mu^2$$

$$X(\mu) = e^{-\frac{x}{\mu}} n \pi \mu + n \pi (-1 + x) \mu \cos[n \pi x] - \frac{1}{2} (-2 + 2 \mu + n^2 \pi^2 (-2 + x) x \mu^2) \sin[n \pi x]$$